NUSHRAT NAUSHIN

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SUMMARY

Materials scientist and metallurgical engineer with 4+ years in synchrotron X-ray spectroscopy and microscopy, including two years as an Advanced Light Source Doctoral Fellow at Lawrence Berkeley National Laboratory, plus 2.5 years of full-time undergraduate teaching and instructional laboratory development. Designs and runs synchrotron experiments (XMCD, XMLD, XPEEM) on thin films and complex alloys, builds statistically rigorous Python analysis pipelines for large experimental datasets, and brings hands-on thin-film deposition, UHV maintenance, and classical mechanical and metallurgical testing.

EXPERIENCE

Advanced Light Source, Lawrence Berkeley National Laboratory ***Sep 2024 – Aug 2026***
ALS Doctoral Fellow

- Independently operate beamlines 4.0.2, 6.3.1.1, and 11.0.1.1; train new users on XMCD and XPEEM measurements.
- Lead 20+ beamtime experiments end to end: experimental planning, in-situ decisions during acquisition, and post-experiment data validation.
- Built XMCD sum-rule analysis with six-source Monte Carlo error propagation, and XPEEM asymmetry-map pipelines (drift correction, MCP noise removal, cross-registration, domain heterogeneity metrics) across temperature series.
- Developed and characterized reference multilayer thin films to improve beamline calibration and reproducibility.
- Co-authored 5+ successful beamtime proposals and contributed to multiple manuscripts.

University of California, Davis ***Sep 2021 – Expected Sep 2026***
Graduate Student Researcher

- Deposit multi-principal element alloy thin films by DC magnetron sputtering and co-sputtering (AJA Orion 8); commission power supplies and maintain UHV systems, including turbomolecular pump repair.
- Lab superuser for the DC/RF sputtering system and Vibrating Sample Magnetometer; trained roughly 20 students on safe instrument operation.
- Correlate structure, composition, and magnetic behavior by combining magnetometry, XRD, and element-specific synchrotron spectroscopy.
- Lead laboratory safety programs, SOP development, and hazardous waste management; recipient of the College of Engineering Safety Excellence Award.

Khulna University of Engineering and Technology ***Jan 2019 – Aug 2021***
Lecturer, Materials Science and Engineering

- Led outcome-based curriculum development and instruction for 8+ undergraduate theory and laboratory courses, typically about 60 students per class.
- Established and upgraded foundry, ceramics, electrochemical analysis, polymer and composites, and heat-treatment instructional laboratories, including equipment specifications and technical tender documents for major purchases.
- Mentored 10+ students through successful graduate school applications.

EDUCATION

University of California, Davis — Ph.D., Materials Science and Engineering ***Sep 2021 – Expected Sep 2026***

- Element-specific magnetism and magnetic domain structure in multi-principal element alloy thin films; dissertation on composition-dependent magnetism in MnCoNi. Advanced to candidacy October 2023. PI: Prof. Roopali Kukreja.

Bangladesh University of Engineering and Technology — B.S., Materials and Metallurgical Engineering ***Oct 2018***

- Honours; GPA 3.79/4.00, ranked 6th of 52. Coursework across extractive and process metallurgy, casting, forming, and joining.

TECHNICAL SKILLS

Thin-Film Synthesis: DC/RF magnetron sputtering and co-sputtering (AJA Orion 8), sol–gel synthesis; UHV operation, maintenance, and troubleshooting including turbomolecular pumps.

Synchrotron and Characterization: XAS, XMCD, XMLD, XPEEM, XPCS, micro-/nano-diffraction; XRR, XRD, VSM, SEM/EDS, AFM, MFM, TEM/electron diffraction.

Data Analysis: Python (NumPy, Pandas, Jupyter, Matplotlib); Monte Carlo error propagation, image registration and similarity metrics; LabVIEW, OriginPro, GenX, Gwyddion.

Mechanical and Metallurgical Testing: UTM tensile testing; Rockwell and Brinell hardness; Charpy and Izod impact; optical emission spectroscopy; metallography; failure analysis.

Tools: Git, LaTeX, AutoCAD, SolidWorks, Inkscape.

SELECTED PUBLICATIONS

- N. Naushin *et al.*, "Element-specific magnetic moment and domain structure in magnetron-sputtered MnCoNi thin films," *J. Magn. Magn. Mater.* (2026). 10.1016/j.jmmm.2026.174250
- W. Beeson, N. Naushin *et al.*, *Adv. Funct. Mater.* (2025). 10.1002/adfm.202424741
- N. Z. Hagstrom, N. Naushin *et al.*, *Nature Communications* (submitted, 2025).
- P. Sharma, N. Naushin, S. Rohila, and A. Tiwari, "Magnesium Containing High Entropy Alloys," in *Magnesium Alloys — Structure and Properties* (2022). 10.5772/intechopen.98557

AWARDS AND PRESS

- ALS Doctoral Fellowship in Residence, Lawrence Berkeley National Laboratory (2024, 2025).
- UC Davis College of Engineering Graduate Excellence Award for Safety (2025).
- BUET Dean's Choice Award (2018); University Merit Scholarship (2015, 2017).
- LPS Qubit Collaboratory Quantum Computing Summer Short Course, Algorithms to Hardware (Jul–Aug 2025).
- Featured in "Chaos by Design" by N. Tamura, ALS (2026); "College Honors MSE Graduate Students for Excellence in Safety," UC Davis (2025); and "International Women's Day Spotlight," UC Davis (2024).

CONFERENCES

- APS March Meeting — Presenter (2025, 2026).
- ALS User Meeting — Invited Speaker (2024); Poster Presenter (2026).

SERVICE AND MENTORSHIP

- Trained roughly 20 students on safe operation of the sputtering system and VSM as lab superuser, and mentored 14+ undergraduate and graduate students on advanced instrumentation.
- Mentored students from underrepresented backgrounds through the UC–HBCU and MESA initiatives, including intensive summer mentorship of a visiting undergraduate researcher.
- Advised 10+ former students on graduate school applications; all are now enrolled in graduate programs.
- Secretary, BUET Material Advantage Chapter: organized an inter-university materials poster symposium (50+ teams) and a STEM competition (500+ participants).
- Member, American Physical Society and The Minerals, Metals & Materials Society.